Lab 7: Electronic Door Lock on FPGA

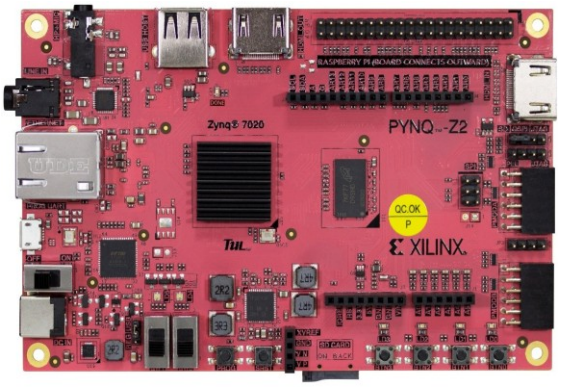
Today, we will put the electronic door lock and keypad encoder from last times onto the FPGA

Discussion – Lab 6

1. Any issues?
2. Solutions to Lab 6

Demonstration Programming FPGA

1. Target Board and Board files
2. Constraints
3. Synthesis, implementation, bitstream generation
4. Post synthesis and post implementation Simulation
5. Other Tricks (Clocking Wizard, Constraints after implementation)



Task – Electronic Door Lock on FPGA

1. If you haven't finished lab 6, work on lab 6
2. Prepare your EDL circuit from lab 6 for FPGA implementation
   a. Implement a clock divider to slow down the system clock
   b. Use LEDs for visual feedback
3. Connect your keypad to the PMOD connector on the PYNQ board
4. Set up your constraints
5. Synthesize, implement, generate bitstream, program board